Alex Morehead

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Machine Learning • Deep Learning • Computational Biology • Generative Modeling

Education

University of Missouri Columbia, Missouri
Ph.D. Computer Science Aug 2020 - Present
- *Dissertation Advisor*: Prof. Jianlin Cheng
- *Cumulative GPA*: 4.0/4.0
- *Relevant Coursework*: Computational Intelligence, Machine Learning/Pattern Recognition, Deep Learning, NLP, Computer Vision, Unsupervised Learning, Computational Systems Biology, Design & Analysis of Algorithms

Missouri Western State University St. Joseph, Missouri
B.S. Computer Science Aug 2016 - May 2020
- *Minor*: Mathematics
- *Cumulative GPA*: 4.0/4.0 with General Studies Honors

Professional Experience

University of Missouri Bioinformatics & Machine Learning Lab Columbia, Missouri
Graduate Research Assistant | Advisor: Jianlin Cheng Aug 2020 – Present
- Research and develop novel geometric deep learning and generative models applicable to computational biology
- Authored and published at ICLR 2022 the Geometric Transformer architecture for protein representation learning
- Published the Geometry-Complete Perceptron Network and Diffusion Model for 3D biomolecular graphs

Profluent AI Research Berkeley, California
Research Intern, Deep Learning | Manager: Jeffrey Ruffolo May 2023 – Aug 2023
- Researched, developed, and evaluated new generative deep learning algorithms for protein design

University of Missouri Allen Angel Capital Education Program Columbia, Missouri
Investment Analyst | Advisor: William (W.D) Allen Aug 2022 – May 2023
- Cultivated deal flow for identifying promising technology start-ups
- Pitched several technology companies to the fund, a subset of which have made it to due diligence
- Ultimately invested in industry-leading blue tech startup technology

Absci AI Research Lab New York City, New York
Research Intern, Deep Learning | Manager: Joshua Meier Jun 2022 – Apr 2023
- Researched and built new deep learning algorithms for antibody design
- Constructed and benchmarked new datasets and deep learning models for proteins
- Collaborated with team members to establish new state-of-the-art results in de novo antibody design

Altec Information Services Saint Joseph, Missouri
Software Development Intern | Advisor: Dan White Aug 2018 – Aug 2020
- Reduced miscommunication between service centers globally by engineering over 5 new Angular web applications
- Built more than 6 secure backend APIs with the Spring framework in an agile development setting
- Collaborated closely with business stakeholders and analysts to understand data and problems needing to be solved

NSF Research Experience for Undergraduates in Data Science of Risk and Human Activity Indianapolis, Indiana
Undergraduate Research Assistant | Program Directors: George Mohler, Mohammad Al Hasan Jun 2019 - Aug 2019
- Authored an ensemble pipeline of 3 convolutional neural networks trained to detect gunshot sounds in the vicinity
- Deployed the pipeline to a cluster of Raspberry Pi 3 microcomputers to evaluate its real-world performance
- Published as a manuscript and orally presented the project's results at IEEE Big Data (2019)

NSF Research Experience for Undergraduates in Synthetic Biology St. Joseph, Missouri
Undergraduate Research Assistant | Program Directors: Todd Eckdahl, Jeffrey Poet May 2018 - Aug 2018
- Investigated applications of combinatorial optimization to model lab experiments performed by synthetic biologists
- Produced Variant Sampler, a Java application for modeling the sample space of in vitro experiments
- Published results of the Variant Sampler project in the American Journal of Undergraduate Research

Publications

• *Accepted/Published* •

Semi-Supervised Graph Learning Meets Dimensionality Reduction — IEEE ICMLA '23 (Virtual)
A. Morehead*, W. Chantapakul*, J. Cheng

DIPS-Plus: The Enhanced Database of Interacting Protein Structures for Interface Prediction — Scientific Data '23 Journal
A. Morehead, C. Chen, A. Sedova, J. Cheng

A Gated Graph Transformer for Protein Complex Structure Quality Assessment and its Performance in CASP15 — Bioinformatics & *ISMB* '23 Lyon, France
X. Chen*, **A. Morehead***, J. Liu, J. Cheng

DRLComplex: Reconstruction of Protein Quaternary Structures using Deep Reinforcement Learning — ICIBM '23 Tampa, FL
E. Soltanikazemi, R. S. Roy, F. Quadir, N. Giri, **A. Morehead**, J. Cheng

Geometry-Complete Diffusion for 3D Molecule Generation and Optimization — ICLR '23 – MLDD Workshop (Virtual)
A. Morehead, J. Cheng

Geometry-Complete Perceptron Networks for 3D Molecular Graphs — AAAI '23 – Graph Deep Learning Workshop Washington D.C.
A. Morehead, J. Cheng

3D-Equivariant Graph Neural Networks for Protein Model Quality Assessment — Bioinformatics '23 Journal
C. Chen, X. Chen, **A. Morehead**, T. Wu, J. Cheng

A Region-Based Deep Learning Approach to Automated Retail Checkout — CVPR '22 – AI City Workshop New Orleans, LA
M. Shoman, A. Aboah, **A. Morehead**, Y. Duan, A. Daud, Y. Adu-Gyamfi

Geometric Transformers for Protein Interface Contact Prediction — *ICLR* '22 (Virtual)
A. Morehead, C. Chen, J. Cheng

High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function — IEEE SuperComputing '21 St. Louis, MO
M. Gao, P. L. Andersen, **A. Morehead**, S. Mahmud, C. Chen, X. Chen, N. Giri, R. Roy, F. Quadir, T. C. Effler, R. Prout, S. Abraham, W. Elwasif, J. Skolnick, J. Cheng, A. Sedova

Synthetic Biology Bicistronic Designs Support Gene Expression Equally Well in vitro and in vivo — AJUR '20 Journal
O. Koucky, J. Wagner, S. Aguilera, B. Bashaw, Q. Chen, A. Eckdahl, E. Edman, P. Gomez, N. Hanlan, N. Kempf, D. Mattoon, S. McKlin, C. Mazariegos, **A. Morehead**, S. Q. Ong, A. Peterson, M. Rojas, K. Roland, K. Schildknecht, H. Seligmann, K. Slater, A. Tauchen, R. Tittor, T. Travieso, D. Urban, C. Willis, J. Zhou, N. L. Snyder, L. J. Heyer, J. L. Poet, T. T. Eckdahl, A. M. Campbell

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IEEE BigData '19 Los Angeles, CA
A. Morehead, L. Ogden, G. Magee, R. Hosler, B. White, G. Mohler

• *Under Review* •

Evaluating Representation Learning on the Protein Structure Universe — OpenReview '23 Under Review
A. Jamasb*, **A. Morehead***, Z. Zhang*, C. Joshi*, K. Didi, S. Mathis, C. Harris, J. Tang, J. Cheng, P. Lio, T. Blundell

• *Preprint* •

EGR: Equivariant Graph Refinement and Assessment of 3D Protein Complex Structures — arXiv '22 Under Review
A. Morehead, X. Chen, T. Wu, J. Liu, J. Cheng

Presentations

Semi-Supervised Graph Learning Meets Dimensionality Reduction — IEEE ICMLA '23 - Oral (Virtual)
A. Morehead

A Gated Graph Transformer for Protein Complex Structure Quality Assessment — ISMB '23 – 3DSIG - Oral
A. Morehead — Lyon, France

Geometry-Complete Diffusion for 3D Molecule Generation — ICLR '23 – MLDD Workshop - Poster
A. Morehead — (Virtual)

Geometry-Complete Perceptron Networks for 3D Molecular Graphs — AAAI '23 – DLG Workshop - Poster
A. Morehead — Washington, D.C.

Geometry-Complete Perceptron Networks for 3D Molecular Graphs — AAAI '23 – AI2ASE Workshop - Oral
A. Morehead — Washington, D.C.

Neural Diffusion Models: Next-Generation Generative Deep Learning — University of Missouri – Deep Learning F22
A. Morehead — Columbia, Missouri

A Region-Based Deep Learning Approach to Automated Retail Checkout — CVPR '22 – AI City Workshop - Oral
A. Morehead, M. Shoman, A. Aboah — New Orleans, Lousiana

Geometric Transformers for Protein Interface Contact Prediction — ICLR '22 - Poster
A. Morehead — (Virtual)

Geometric Transformers for Protein Interface Contact Prediction — Shandong University '22 - Oral
A. Morehead — Qingdao, Shandong Province, China (Virtual)

High-Performance Toolbox for Deep Learning of Protein Structure and Function — IEEE SuperComputing '21 - Oral
A. Morehead, A. Sedova, M. Gao — St. Louis, Missouri (Virtual)

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IEEE BigData '19 - Oral
A. Morehead, L. Ogden, G. Magee — Los Angeles, California

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IUPUI Student Summer Symposium - Poster
A. Morehead, L. Ogden, G. Magee — Indianapolis, Indiana

Variant Sampling in vitro with a Scheduling Twist — Alpha Chi National Convention '19 - Poster
A. Morehead — Cleveland, Ohio

Variant Sampling in vitro with a Scheduling Twist — MWSU PORTAL Summer Research Showcase '18 - Poster
A. Morehead — St. Joseph, Missouri

Predicting Game Genres by Analyzing Code Structure — CSCC Central Plains Conference '18 - Poster
S. Frazier, **A. Morehead**, S. Prine — Maryville, Missouri

Predicting Game Genres by Analyzing Code Structure — MWSU Multidisciplinary Research Day '18 - Poster
S. Frazier, **A. Morehead**, S. Prine — St. Joseph, Missouri

Leadership Activities

Reviewer for the NeurIPS AI4Science and GenBio Workshops	Sep 2023 – Present
Reviewer for the Learning on Graphs (LoG) Conference	Sep 2023 – Present
Reviewer for the ICML Workshop on Computational Biology (WCB)	Jun 2023 – Present
Reviewer for IEEE Transactions on Neural Networks and Learning Systems (TNNLS)	Apr 2023 – Present
Reviewer for Nature Machine Intelligence	Feb 2023 – Present
Reviewer for IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI)	Feb 2023 – Present
Volunteer for the International Conference on Learning Representations (ICLR)	Apr 2022 – Apr 2022
Reviewer for Neural Information Processing Systems (NeurIPS)	Sep 2021 – Present
Upsilon Pi Epsilon (ΥΠΕ) | University of Missouri Chapter	Aug 2020 - Present
Alpha Chi (AX) | Missouri Western State University Chapter	Mar 2018 – May 2020
Kappa Mu Epsilon (KME) | Missouri Western State University Chapter	Mar 2018 – May 2020

Awards & Grants

CVPR Travel Grant	*Jun 2022*
XSEDE Startup Hardware Grant	*Dec 2021*

Dean's Engineering Excellence Fellowship	*Aug 2020*
James W. and Joan M. O'Neill Graduate Fellowship in Engineering	*Aug 2020*
MWSU Outstanding Graduating Computer Science Student Award	*May 2020*
MWSU President's Honor Roll	*May 2020*
Floyd Tesmer/Strayer University Prize in Computer Science and Engineering	*Apr 2019*
Alpha Chi Region IV Scholarship	*Apr 2019*
Grand Midwest Asynchronous Programming Contest 3rd Place Prize	*Apr 2017*
East Side Lions Club Scholarship	*May 2016*

Projects

PyTorch Lightning (20k+ Stars) | GitHub: https://github.com/Lightning-AI/lightning Oct 2022 – Present

- Work as a Lightning League Ambassador to ensure the quality of PyTorch Lightning code

Lightning-Hydra (3k+ Stars) | GitHub: https://github.com/ashleve/lightning-hydra-template Jan 2023 – Present

- Collaborate with other machine learning engineers to offer a first-class open-source deep learning project template
- Contributed generic learning rate scheduling and optimization configurations to the project

BioPython (3k+ Stars) | GitHub: https://github.com/biopython/biopython Apr 2022 – Present

- Created functionality to remove large feature generation temporary files after all processing completes
- Collaborated with senior project developers to ensure proposed solution matched existing code standards

Graphein (800+ Stars) | GitHub: https://github.com/a-r-j/graphein Apr 2022 – Present

- Added workflow to catch elusive OS errors when executing external programs
- Worked with senior project developer to ensure workflow matched existing code standards

EGNN-PyTorch (150+ Stars) | GitHub: https://github.com/lucidrains/egnn-pytorch Apr 2021 – Present

- Noticed the omission of an important geometric unit test for this neural network architecture
- Developed and validated the code to test for node-wise permutation equivariance within the network

AMG-List (180+ Stars) | GitHub: https://github.com/amorehead/awesome-molecular-generation July 2022 – Present

- Curated an open-source list of machine learning research papers related to molecular generation
- Through pull requests, collaborate with other researchers to maintain the quality of papers presented

Bio-Diffusion (80+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/bio-diffusion Feb 2023 – Present

- Created a state-of-the-art open-source molecular diffusion model for 3D biomolecule generation
- Maintain code quality by resolving new issues and contributing new code periodically

DeepInteract (45+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/DeepInteract Oct 2021 – Present

- Designed and developed a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts
- Maintain project by addressing bug fixes, feature requests, and future planning

Protein Workshop (35+ Stars) | GitHub: https://github.com/a-r-j/ProteinWorkshop Sep 2023 – Present

- Added new geometric/sequence-based neural network architectures to benchmark protein representation learning
- Maintain project by contributing new datasets and emerging model architectures occasionally

Technical Skills

Programming: Python 3 • Java • Angular 2+ • C/C++ • HTML/CSS • SQL • C# • R • MATLAB

Tools/Frameworks: PyTorch • TensorFlow • NumPy • Pandas • Spring Boot • NodeJS • Docker • AWS • Google Cloud

Methodologies: RESTFUL APIs • Version Control {Git} • Agile Development

News & Media Outreach

[1] MU Engineering. (2020, October 27). *Alex Morehead*. University of Missouri College of Engineering. https://engineering.missouri.edu/alex-morehead/.

[2] MU Engineering. (2021, October 18). *New protein prediction tool could accelerate biological discoveries*. University of Missouri College of Engineering. https://engineering.missouri.edu/new-protein-prediction-tool-could-accelerate-biological-discoveries/.

[3] ORNL. (2021, January 10). *Summit Powers Novel Protein Function Prediction Work*. Oak Ridge National Laboratory. https://www.ornl.gov/news/scientists-use-summit-supercomputer-deep-learning-predict-protein-functions-genome-scale/.

[4] HPCwire. (2021, January 13). *Summit Powers Novel Protein Function Prediction Work*. HPCwire. https://www.hpcwire.com/2022/01/13/summit-powers-novel-protein-function-prediction-work/.

[5] MarkTechPost. (2022, January 18). *Researchers Introduce High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function*. MarkTechPost Tech News – AI Paper Summary. https://www.marktechpost.com/2022/01/18/researchers-introduce-high-performance-deep-learning-toolbox-for-genome-scale-prediction-of-protein-structure-and-function/.

[6] Show Me Mizzou. (2023, February 8). *Artificial intelligence, real ambition*. University of Missouri. https://showme.missouri.edu/2023/artificial-intelligence-real-ambition/.